



Navigating procurement strategies adoption in the New Space transition: Drivers and conditions

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ABSTRACT

The space industry is undergoing significant transformations. Increased private sector participation, new satellite-based services, reduced launch costs, digitalisation, and innovative space infrastructure designs decrease entry barriers. New Space companies from the private sector introduce innovations, spin-ins, capabilities, expertise, and practices. At the same time, space agencies contend with time constraints and the imperative to ensure value for money in times of constrained budgets.

In this dynamic environment, space agencies must evolve approaches and interactions with the private sector to reaffirm their pioneering role in space exploration, research, and commercialisation. As many New Space organisations are still in their growth phase, space agencies aim at supporting the New Space ecosystem by fostering competition, facilitating capacity building, and catalysing market creation. In parallel, expanding the pool of bidders enables space agencies to avoid being locked into potentially inefficient solutions when acting as client. As space agencies transition towards the New Space paradigm and intensify collaborative endeavours, there exists a knowledge gap concerning the optimal utilisation of conventional approaches, such as cost-plus contracts, versus commercially driven practices like service contracts and parallel bidding. This paper aims to investigate the drivers and conditions shaping procurement choices in the space sector, their influence on project outcomes, and the implications for space agencies and policymakers when making procurement decisions and designing PPPs for future programmes or projects.

Following qualitative research, we conducted a multiple case study focusing on projects within space agencies governed by traditional and New Space strategies. Employing a polar case selection approach, we analysed successful and challenging instances stemming from the implemented procurement practices in 23 programmes/projects led by Western and non-Western space agencies. By leveraging primary and secondary data sources, our investigation illuminates the motivations, best practices, and limitations associated with the conditions of traditional and New Space contexts.

We provide a set of key drivers and conditions guiding the configuration of procurement practices within space agencies and their adoption. By comparing literature on procurement practices, both in space and non-space domains, with empirical evidence, our analysis reveals that a "one-size-fits-all" approach does not exist. Each PPP prioritizes distinct objectives (e.g. early technology development, demonstrators, first operational hardware, cost reduction, commercial purposes), and accommodates for varying levels of risk appetite. The choice of approach is therefore determined by the interplay between objectives and associated risks. Furthermore, our results confirm the role of drivers and conditions highlighted in non-space PPP literature, while introducing new ones that are peculiar of the space domain. Finally, we unfold several interplays between drivers and conditions leading to procurement choices that not necessarily aligns with the precepts of general PPP literature. Main results are condensed in five propositions.

Our contribution to practice encompasses a set of drivers and conditions facilitating the selection of procurement practices by space agencies. By incorporating considerations of both space agencies and bidding companies, and by highlighting the interplay of drivers and conditions guiding the choice of procurement practices, our research supports space agencies in the transition from a pure traditional paradigm to a mixed one, Traditional and New Space, where commercially driven initiatives are also undertaken when appropriate.

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1. Introduction

As the space sector evolves in response to changing market dynamics, the management of legacy space activities differs from the approach favoured in the current New Space era. Historically, space agencies like NASA and Roscosmos dominated space activities through command-and-control and hands-on approaches [1]. Space agencies were responsible for developing industrial capabilities and expertise from scratch and managing government funds to achieve strategic objectives. They maintained therefore stringent control over system requirements to ensure high performance of space assets and the efficient use of public funds. The high-risk nature of the space sector demanded strict oversight efforts, with the public sector assuming the risks and costs associated with asset development [2]. This scenario dominated until the emergence of the New Space in the early 2000s [3], characterized by the entry of private companies from both space and non-space sectors [4]. Barriers to entry decreased, and private companies stimulated the introduction of innovative products and services [5,6]. Companies and investors start recognizing the commercial potential of space activities, stepping in as complementary source to public funding.

In the legacy space paradigm, space agencies developed core capabilities and competencies to support a previously non-existent market and industry. This was achieved through aligned procurement practices and strategies (e.g., command-and-control, hands-on approaches, and cost-plus contracts) [1]. These practices laid the foundation of today's space sector in many countries but focused on industrial capacity building rather than cost-efficiency. To leverage the value and contributions from the private sector, space agencies started adapting their practices to align with the New Space paradigm [7]. On the one hand, space agencies refrain from imposing all requirements and system specifications, as such rigidity hinder the innovative spirit of the private sector. On the other hand, contributions from private investors help balance decreasing public funds, provided that a viable commercial market for space-based products and services is present [2]. Many space segments, such as in-orbit servicing, remain immature, necessitating tailored public sector intervention for technologies to surpass the initial and most significant levels of technology risk. The New Space era results therefore in a market where private intervention is encouraged in various projects, necessitating the space agency to step back and prioritize insights over oversight during the development process [3,8]. Concurrently, immature and high-risk technologies still require a nuanced approach to risk and cost-sharing between public and private entities. Innovation, resource sharing, and accelerated progress might unfold when the public and private sectors collaborate through various forms of partnerships. Indeed, there is a growing prevalence of Public-Private Partnerships (PPPs) in the space sector [2]. Nevertheless, the readiness to adopt this evolving paradigm varies significantly across space market segments and institutional contexts.

In academic research, a clear and unequivocal framework outlining best practices for procurement among space agencies is lacking [2]. Concurrently, there is ambiguity regarding the criteria that should guide space agencies in supporting innovative procurement practices (e.g. pre-commercial procurement, forward-buying approaches) as opposed to traditional ones (e.g. cost-plus contracts), alongside empirical evidence supporting commercial over conventional practices [9].

In the space sector, it is impractical to establish a universal procurement framework that applies uniformly across all contexts, space agencies, and industrial ecosystems; projects have distinct characteristics, space agencies and companies have divergent objectives and priorities, market and industrial bases display distinct features, and space agencies dispose of different budgets and capabilities. Although there is growing discussion on New Space procurement and space PPPs [2,3,8], there is limited knowledge about the conditions necessary for their successful implementation in space projects and missions. What is lacking is an evidence-based understanding of when and why specific procurement practices succeed, and how PPPs can be shaped to balance

technological risk with market opportunity. This lack of clarity may lead to a precipitous shift towards New Space practices (e.g. pure service contracts), potentially resulting in inefficient resource allocation and utilisation, as well as adverse effects on national space industries. Nuanced practices between legacy and New Space approaches may facilitate a more gradual transition and, if necessary, a reversion towards alternative practices.

Against this backdrop, this research aims to identify the drivers and conditions under which space agencies and private actors adopt different procurement models, to understand how these choices impact project outcomes, and to generate lessons for space agencies and policymakers facing similar trade-offs.

We address this objective through an exploratory, abductive research design. We conducted a multiple case study on 23 space projects, utilizing both primary data (semi-structured interviews) and secondary data (academic records, reports, websites, newspapers). The approach combined exploration of case dynamics with iterative engagement with literature, allowing emerging patterns from data to be confronted with and informed by existing conceptual frameworks. We performed both within-case and cross-case comparisons to identify key drivers and conditions guiding the selection of procurement practices and to gather empirical evidence on the relationship between these drivers and conditions and the selected practices.

We compare drivers and conditions consistent with non-space literature and highlight the peculiar ones to the space domain. We examine how the joint effect of these drivers and conditions shapes procurement practices in the space sector.

The remainder of the paper is structured as follows. Section 2 provides an overview of PPP literature and recent academic discourse on space PPP typologies. Section 3 delineates the qualitative methodology implemented. Section 4 presents the empirical findings, and Section 5 discusses them against the literature and puts forth propositions as a conceptual synthesis of the core original contribution of the study. Section 6 offers concluding remarks, implications, limitations of the paper, and avenues for future research.

2. Background knowledge

2.1. Traditional Public-Private Partnerships (PPPs)

Public-private partnerships (PPPs) are long-term agreements between a public sector entity, e.g. the government or one of its agencies, and partners from the private sector [10]. In a PPP, the private sector provides assets and services conventionally provided by public entities; part of the associated risk and management responsibility is thereby transferred from the public to the private partner [11,12]. The compensation to the private entity is contingent upon the achieved performance [11]. PPPs have been extensively utilized for the provision of infrastructures such as hospitals, prisons, railways, schools, and roads [10,12], especially to support development in emerging economies [13].

For the successful engagement of private partners, PPPs necessitate well-defined supporting frameworks and careful planning [13]. The public entity must establish clear objectives and expected outcomes of the PPP, alongside risk management and assessment procedures [13]. The PPP should be tailored to the local context and the specific requirements of each sector [13], and periodic contract reviews should facilitate adjustments to accommodate evolving market conditions and dynamics [10]. Indeed, PPP contracts must possess sufficient flexibility to allow for term modifications based on performance monitoring [13]. A PPP needs to be periodically evaluated to ensure it remains the most efficient framework for procuring the intended assets and services [10]. Primarily, competition and a competitive procurement process are essential for PPPs to enhance the likelihood of efficiency, cost-effectiveness, and mutual benefit for both sectors [10,13]. Absent these conditions, PPPs could lead to monopolistic scenarios where a single private entity dominates the market, increasing entry barriers for

other private actors [10].

PPPs bundle together various phases and functions of a project, with the responsibilities assigned to the private partner contingent upon the asset or service being provided [11]. Typically, the functions integrated within these partnerships encompass the design, building, financing, maintenance, and operational stages [11]. Classification of various types of PPPs is based on the extent to which traditionally public tasks, risks, and responsibilities are transferred to a private partner [14–17]. The most common categories of PPPs differentiate between Operation, Maintenance, and Service contracts, such as those implemented to enhance efficiency, with the public sector providing capital investments and monitoring performance [14,18]. Additionally, within Build, Operate, and Invest arrangements, private partners assume financial responsibilities, with the regulatory framework for managing projects and contracts becoming more complex [14]. Furthermore, there are State-owned Enterprises and Joint Ventures, in which the realisation and/or operation of infrastructure projects is managed by a Special Purpose Vehicle (SPV) [14,19]. In these arrangements, the government holds the shares, yet the entity operates in a businesslike manner [14]. These approaches diverge from conventional public procurement, such as the Design-Bid-Build model, where design and construction are awarded separately and sequentially to private firms [20]. Conversely, at the other end of the spectrum, full privatisation involves arrangements such as Build-Own-Operate, where the contractor retains ownership of the asset, assumes operating revenue risk, and benefits from surplus revenue [20]. Another example of full privatisation is asset sale, in which a previously publicly financed asset is transferred to a private entity indefinitely [20]. Table 1 summarizes traditional PPP categories.

PPPs should be prioritized when their benefits are justified, particularly when Value for Money (VfM) (i.e. the project is completed at a lower cost or with improved quality for the same expenditure, [14]) is improved compared to traditional public procurement methods [11]. This necessitates a thorough evaluation to determine if projects satisfy the criteria for PPP consideration, as well as careful planning and design of the PPP framework.

2.2. Space Public-Private Partnerships (PPPs)

The advent of commercial space activities related to space exploration, satellite manufacturing and service-related businesses, and space tourism, enhance risk and cost-sharing efforts between public and private entities [21], marking a departure from traditional procurement models based on direct infrastructure acquisition. Legacy PPP arrangements in space were primarily structured around state-led investments and bespoke developments, favouring large incumbents and minimizing private sector risk [8]. In contrast, contemporary PPPs adopt market-oriented procurement practices, characterised by service-based

contracting, milestone-based funding, and commercial incentives [2]. This transformation is evident when examining large-scale programs like NASA Artemis, which blends different procurement approaches and governance models. Legacy components (e.g. SLS and Orion) continue to operate under cost-plus contracts, while more recently introduced assets, such as the Lunar Gateway, the HLS, and its associated logistics services, are developed through PPPs that emphasize performance-based incentives and shared investment structures [9]. Such hybrid procurement strategies are emblematic of the “New Space” economy, where competition, commercial responsiveness, and operational risk-sharing are prioritized [22]. This shift is not merely procedural but reflects a broader structural evolution; new PPP frameworks institutionalize co-development and commercial off-the-shelf (COTS) integration as part of long-term procurement strategies [23].

In the space sector, this reflects a trend from conventional procurement practices, rooted in space agency ownership and control, toward hybrid, service-based practices, with varying degrees of public and private sector involvement [2]. The main contractual frameworks employed in the space sector vary across the following categories. Contractual partnerships represent the conventional contractual agreements under which the contracted private entity provides space products and services to the public entity [2,24]. Collaborative partnerships and projects, where PPPs fall, are joint ventures and alliances between public entities (i.e. space agencies) and private firms where the sharing of risks, investments, and resources promotes technological advancements and rewards for both sectors [2,21,24,25]. These entities may give rise to consortia, i.e. groups of companies and public institutions jointly working on large-scale programs and projects [24]. In an attempt to categorize how PPPs unfold and adapt in the current New Space scenario, Kim (2023) [2] stresses the nuanced and hybrid transition of PPPs from a high degree of public involvement, towards a higher weight of private presence. In his Operation Concession (OCon) typology, the role of the public entity dominates over the private one, with the public entity financing, designing, developing, and transferring the space asset to the private entity, merely tasked with its operation. This typology resembles the Operation, Maintenance, and Service contracts in general PPPs [2,14,18]. Gradually shifting towards an increased private participation, the Partially Finance-Design-Develop (PFD) typology resembles the Build-Finance contracts reported in general PPP literature (without operation), with parallel finance, design, develop tasks assigned to both the private and public entities, or with the private entity bearing one or two of these tasks. Here, the space infrastructure is owned and fully operated by the public entity. The Partially Finance-Design-Develop and Fully Own-Operate (PFD-FO) typology is similar to the PFD typology, but with the private entity owning and operating the space asset. This typology resembles the full privatisation typology of Build-Own-Operate PPPs [20]. Last, Partially Finance and Fully Design-Develop-Own-Operate (PF- FDO) is presented as PPP with

Table 1
Overview of traditional PPP categories in non-space domains.

CATEGORY	DEFINITION	RISK AND RESPONSIBILITY ALLOCATION	OWNERSHIP AND FINANCING	EXAMPLES
Conventional Procurement (Design-Bid-Build)	Sequential contracting for design and construction	Public bears most risks	Public funds, private executes tasks	Traditional infrastructure projects
Operation, Maintenance, and Service Contracts	Private sector operates/maintains public assets	Private manages service performance, public retains core risks	Public provides capital, private compensated by performance	Hospital management, road maintenance
Build-Operate-Invest	Private sector designs, builds, and finances, then operates asset for a period	Private assumes financial and operational risk	Mixed; private investment and potential public subsidies	Railways
State-Owned Enterprise/ Joint Venture	Mixed-ownership entity created for project delivery	Risks shared between public and private	Government holds shares but entity operates business-like	Airports
Build-Own-Operate	Private builds, owns, operates indefinitely	Private bears financial, operational, and market risk	Fully private ownership	Energy plants
Asset Sale/Full Privatisation	Transfer of existing public asset to private sector	All risks transferred to private	Fully private	Privatisation of utilities

the highest private responsibility, which in PPP literature is defined as Build-Own-Operate [20]. Here, financial resources come from both private and public entities, but design, develop, own, and operate tasks are fully in the hands of the private entity. On the two extremes of the spectrum, we find the Conventional Procurement, with the sole role of the public entity taking care of all phases and bearing all risks, and commercial development, with the private entity managing all phases and bearing all risks. Under the latter category fall forward-buying practices, in which public agencies commit to purchase future services or products in advance, de-risking private investment without assuming ownership [2]. An example is the NASA Commercial Lunar Payload Services (CLPS) program, in which the space agency contracts private companies to deliver payloads to the lunar surface, paying upon successful delivery while companies retain and reuse infrastructure for commercial purposes [26]. Similarly, forward-buying is evident in commercial launch services where agencies commit to block purchases of platforms like Falcon rockets or Starship liners, securing access to delivery services once maturity is demonstrated by contractors. More recently, Takata and Hidaka [23], through an in-depth analysis of partnerships between JAXA and private Japanese companies, have proposed an additional category, referred to as Fully Finance and Fully Design-Develop-Own-Operate with Co-Creation (FF-FDO-C). Positioned between PF-FDO and full commercial development PPPs, in this procurement practice the public sector focuses on national-level R&D and technologically challenging missions, while supporting high-performance private space businesses [23]. As JAXA J-SPARC program exemplifies, the public sector is therefore responsible for developing new technologies in alignment with its own strategic R&D priorities, providing public funds exclusively for public-sector R&D rather than for private business creation [23]. Table 2 provides an overview of space PPP categories, and Fig. 1 highlights their positioning along the space PPP spectrum.

2.3. Impact of geopolitical and strategic autonomy on space procurement choices

Partnerships in space are not new. Historically, international collaborations such as the ISS have demonstrated how agencies pool resources and capabilities to achieve strategic objectives [27]. Beyond established agencies, partnerships serve as capability-building

mechanisms for emerging space nations. For example, emerging regional space agencies such as the African Space Agency, the Arab Space Coordination Group, and the Latin American and Caribbean Space Agency foster cross-country collaboration to strengthen industrial bases and enhance technological sovereignty [28]. These initiatives highlight how partnerships are still the key ingredient for success in space [28, 29].

Today, however, geopolitical tensions and the pursuit of strategic autonomy are giving PPPs renewed prominence, echoing Cold War-era drivers of programs like Apollo. The New Space race is increasingly tied to geopolitical competition, unique military capabilities development, and national prestige [22]. The growing reliance on PPPs and service-based contracts in space is not driven solely by New Space entrants but also by shifting geopolitics and intensifying competition among the U.S., China, and Russia [30]. Partnerships allow governments to tie private successes to national prestige; by engaging start-ups and established firms, states both validate private actors as capable partners and signal national technological leadership [30]. For companies, association with flagship national programs strengthens their reputation, visibility, and financial attractiveness, while for agencies it reframes collaboration as a coalition of national excellence [30]. This pattern is visible across major space powers. In the U.S., PPPs support efforts to return first to the Moon and maintain global leadership, blending commercial and international partnerships [31,32]. Japan leverages PPPs within programs such as J-SPARC to foster commercialisation and accelerate new technology acquisition, and Japan even reformed the Space Resource Act to strengthen private participation [33]. India, after decades of international collaborations oriented toward socioeconomic development, is now reorienting toward competitiveness in the Asian space race [34]. By mobilizing both small and large firms through PPPs, India aims to address capacity deficits while expanding its military space and dual-use capabilities.

If, on the one hand, geopolitical tensions and the pursuit of strategic autonomy drive governments to strengthen PPPs with national firms [28,30,34,35], the type of project often determines procurement dynamics. Dual-use projects create a key set of incentives. When technologies have potential civil and commercial applications, PPPs allow risks and costs to be shared effectively. The presence of commercial unfolding provides private partners with clear return prospects, encouraging them to co-invest and co-develop. In this sense, dual-use programs strike a

Table 2
Overview of space PPP categories (adapted from Kim, 2023).

Space PPP category	Task allocation			Risks		
Sector	Public	Both	Private	Public	Both	Private
Conventional	Finance Design Operate		Build	Financial Construction Technical Business Operational		
Ocon (Operation Concession)	Finance Design Develop Own		Operate	Financial Construction Technical	Business	Operational
PFD (Partially Finance-Design-Develop)	Operate Own	Finance Design Develop		Business Operational	Financial Construction Technical	
PFD-FO (Partially Finance-Design-Develop and Fully Own-Operate)		Finance Design Develop	Operate Own		Financial Construction Technical	Business Operational
PF-FDO (Partially Finance and Fully Design-Develop-Own-Operate)		Finance	Design Develop Operate Own		Financial Construction	Technical Business Operational
Commercial		Operate	Finance Design Build			Financial Construction Technical Business Operational

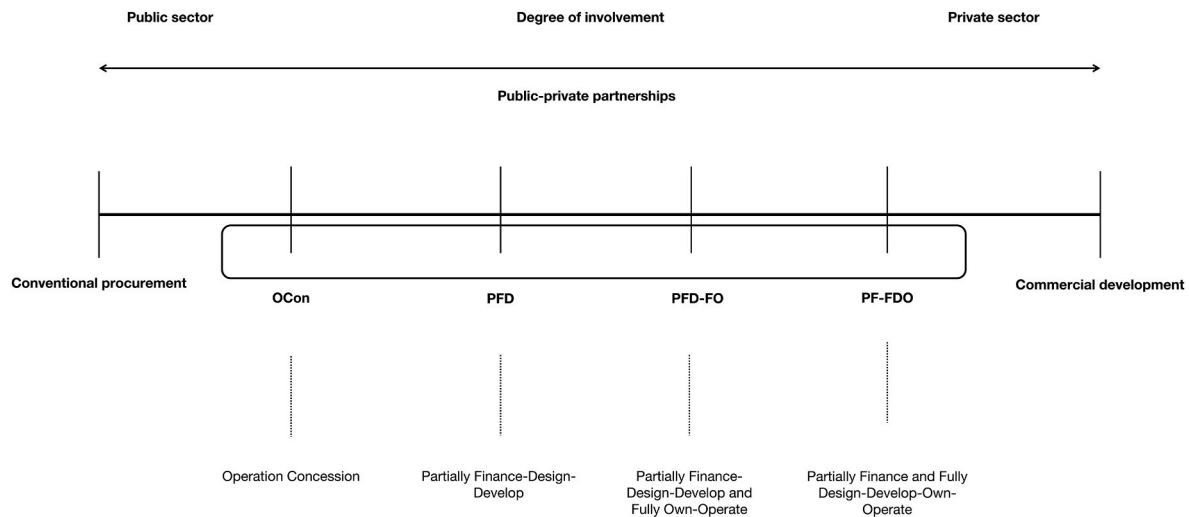


Fig. 1. Overview of space PPP categories along PPP spectrum (adapted from Kim, 2023).

balance, i.e. meeting national security objectives while attracting private firms through incentives tied to commercial markets.

2.4. Key benefits and challenges of space PPPs

Some key benefits foster the adoption of PPPs within the space sector. First, the combination of expertise, know-how, and resources from the public and private sectors accelerate technology development and innovation; space agencies share expertise and data with private entities, resulting in enhanced national capabilities in space-related activities and market segments (e.g., space exploration, satellite communication) [21,24]. Jointly, private companies are incentivized to introduce innovative technologies and efficient solutions with rapid development cycles pushed by the competitive nature of the New Space paradigm [36]. This may result in market expansion towards new applications from space products, assets, and services [24]. A key benefit remains the sharing of costs and risks between the entities engaged in the PPP; more efficient use of resources may unfold, crucial to deal with the challenges posed by the unique risky and uncertain nature of space activities [21,24]. Given the identified benefits, space PPPs hold promise to stimulate and accelerate the New Space growth.

Even if opportunities arise to learn from other sectors implementing PPPs, with the main one being the transportation sector (similar long-term commitment and investments, flexible PPP structures to encourage competitive and innovative environments to adapt to new technologies and challenges, and infrastructure projects translating into public benefit and value for society [36,37]), some peculiarities within the space sector may impose specific challenges in space PPPs implementation. If risk identification, sharing, and mitigation is a critical aspect of PPPs, its management in the space sector may not be aligned with conventional PPP risk allocation methods [2,38]. The inherent risks of design, deployment, operations, and maintenance of space assets are more critical than in other sectors [38], yet risk allocation is a crucial task given the high uncertainties of space activities [25]. Showcasing the public value and benefit stemming from space assets developed with PPP contracts may be challenging, as society is not directly using space infrastructures like in railway and other transportation instances. In space programs and projects, coordination among stakeholders, both public and private, may result in challenges following the scale and complexity of projects, especially when clear governance structures and well-defined roles and responsibilities of stakeholders are lacking [9, 38]. Market and commercial viability of space assets, products, and services may be hard to forecast and ensure, especially in the case of innovative technologies and applications, when markets are emerging

and space agencies seem to remain the sole customer. This may hinder investments from private entities, and limit their willingness to bid for projects, which are crucial pillars of successful PPPs. Last, collaborative culture between public and private entities must be fostered, with collaborative engagement efforts, dialogues, and agreements often across borders, key to ensuring sustainable long-term PPPs [2,21,24]. This may pose challenges and take time in the space sector, especially within entities, both private and public, used to conventional procurement approaches and a close innovation culture [9,39].

3. Methodology

3.1. Data collection and sampling

To address the research objective we favoured an abductive qualitative research design, well-suited to contexts where background knowledge is scattered and phenomena are emergent, under-theorized, and highly context-dependent [40]. The study draws on a comparative multiple case study [41] investigating the emergent practices surrounding PPPs in the space sector. Space PPP arrangements are still under development and their outcomes remain contingent and context-dependent; limited prior theorisation exists on their organisational consequences, optimal conditions of use, performance implications, and many lessons learnt from PPPs in other industries are not entirely applicable to the space sector given its peculiarities. Qualitative, exploratory methods are particularly suited to contexts marked by high novelty and contextual dependence, where the goal is to explore and interpret processes and uncover emerging patterns and concepts as they unfold [41,42]. The multiple case study was selected to find recurrent patterns and insights and to confirm and disconfirm prior knowledge on PPPs adoption, that was largely developed with reference to non-space contexts [43–45].

We implemented both white and grey literature [46] to collect data on 23 space cases. The cases were purposefully selected [47] to cover all the space PPP typologies proposed by Kim (2023) [2]. The dataset of cases includes both programs (e.g., COTS, CLPS, Galileo) and projects (e.g., ICESat-2, SLIM, BEAM). Programs allow space agencies to experiment with different procurement models under a common umbrella, and based on how the program unfolds (e.g., with the SPOT satellites). By contrast, projects are initiatives aimed at delivering a specific mission, system, or infrastructure element. Procurement logics differ across levels; program-level arrangements emphasize portfolio balance, industrial base development, and strategic alignment, whereas project-level practices are more closely tied to TRL advancement,

mission maturity, and risk management. Including both types of initiatives in the sample enabled us to capture how PPP categories operate not only at the level of individual missions but also within broader strategic plans. Each case was mapped to one of the PPP categories proposed by Kim (2023) [2], based on task and risk allocation between public and private partners (as proposed by Kim (2023) [2]), while also considering hybrid characteristics, such as project complexity, firm maturity, evolving requirements, and procurement structures, drawn from PPP and innovation literature (e.g., [10,11,48]). This additional layer enabled us to examine intra and inter-category variation and emerging PPP patterns. In so doing, and following abductive qualitative research [49,50], we combined deductive application of established PPP typologies with inductive comparative exploration of case-specific features, enabling both classification and contextual interpretation. Table 3 presents a summary description of the selected space cases and categorizes them under the corresponding space PPP typology. The selection of cases ceased upon reaching theoretical saturation [51], i.e. when all categories of space PPPs were represented in the sample and no additional cases (based on available secondary data) offered sufficiently complete or relevant information to justify inclusion. Secondary data, i.e. academic records, but also newspapers, handbooks, space agencies' reports, and websites [46], were complemented with primary data, i.e. semi-structured interviews [52]. Semi-structured interviews were conducted with key stakeholders from space agencies and space companies involved in the selected cases. These included decision-makers from space agencies, and, where agencies were inaccessible, representatives from private sector companies involved in the PPPs. We focused on established agencies because they possess a longer track record of PPP implementation, and more documentation of procurement practices. This provided a richer empirical base to analyse the evolution of procurement practices and to capture both conventional and emerging practices. Furthermore, established space agencies often act as global reference points, setting standards that are later adopted or adapted by other space agencies. Informants were purposively selected [53] based on their knowledge of partnerships and collaborations between public and private entities in space programs and projects. A snowball sampling technique enabled to enlarge the sample of experts [54]. The sample comprises 9 informants. Table 4 provides an overview of the sample of interviewed experts.

We employed a standardized semi-structured interview protocol to guide interviews, while retaining flexibility to adapt and refine questions as themes emerged. The protocol covered key areas, such as objectives and context of the implemented PPPs; the rationale behind selected procurement practices; the distribution of roles, risks, and responsibilities between public and private partners; perceived benefits and challenges of the chosen approach; lessons learned and recommendations for future adoption. All interviews were conducted remotely, recorded when the consent of informants was granted (otherwise, with detailed hand notes), and transcribed verbatim for analysis.

3.2. Data analysis

A dataset of the space PPP cases was compiled through desk research, drawing from academic literature, official agency reports, and project documentation. Each case was categorised according to Kim Moon (2023) [2] space PPP taxonomy, based on the defining characteristics outlined in his framework. These categories served as initial structure for comparative analysis. To favour a robust cross-case comparison, we built a comparative framework, aimed at the collection of relevant case characteristics. The framework was developed through a two-phase process. In the deductive phase, it was constructed based on the core characteristics of space and non-space PPPs as suggested in the literature (e.g., contractual design, funding sources). In the inductive phase, it was refined by incorporating case-specific characteristics derived from secondary and primary data from the selected cases ([53,55]). To enhance

Table 3

Case studies and related space PPP categories.

Space PPP category	Case	Brief description
Conventional	<i>Next-Gen OPIR (Next-Generation Overhead Persistent Infrared)</i>	Four satellites providing early warnings of ballistic and hypersonic missile launches targeting the US (U.S. Space Force led)
OCon (Operation Concession)	<i>Galileo (1st gen.)</i>	European global navigation satellite system, interoperable with GPS (US) and GLONASS (Russian) (ESA and European Commission led)
	<i>Davinci (Deep Atmosphere Venus Investigation of Noble gases, Chemistry, and Imaging)</i>	Research project to explore Venus (NASA led)
	<i>SPOT 1-2-3 (French Earth Observation Satellites)</i>	Earth imaging satellites developed by CNES
	<i>Icesat-2 (Ice, Cloud, and land Elevation Satellite)</i>	Satellite to investigate changes in the cryosphere (NASA led)
PFD (Partially Finance-Design-Develop)	<i>Maven (Mars Atmosphere and Volatile EvolutionN)</i>	Mission dedicated to determine the loss of Martian atmosphere over time (NASA led)
	<i>OCO-1, OCO-2 (Orbiting Carbon Observatory)</i>	Science satellites (NASA led)
	<i>ML-2 (Mobile Launcher 2)</i>	Ground platform that will launch the Space Launch System (SLS) (NASA led)
	<i>SLIM (Smart Lander for Investigation Moon)</i>	Mission to demonstrate lunar landing techniques (JAXA led)
	<i>H3</i>	Launch vehicle to launch commercial satellites (JAXA led)
PFD-FO (Partially Finance-Design-Develop and Fully Own-Operate)	<i>BEAM (Bigelow Expandable Activity Module)</i>	Expandable habitat attached to the ISS (NASA led)
	<i>MMX Space Probe (Martian Moons eXploration Mission)</i>	Spacecraft for remote sensing and observations of both moons (JAXA led)
	<i>MRV (RSGS) (Mission Robotic Vehicle, part of Robotic Servicing of Geosynchronous Satellites)</i>	Experimental satellite servicing vehicle (DARPA led)
PF-FDO (Partially Finance and Fully Design-Develop-Own-Operate)	<i>CRD2-Phase 1 (Commercial Removal of Debris Demonstration)</i>	Phase 1 aimed at controlling space debris (JAXA led)
	<i>COTS SpaceX (Commercial Orbital Transportation Services)</i>	Program for the development of private spacecraft and launch vehicle (NASA led)
	<i>COTS Orbital Sciences (Commercial Orbital Transportation Services)</i>	Program for the development of private spacecraft and launch vehicle (NASA led)
	<i>Gateway Logistics Services (GLS)</i>	Unmanned space flights to the Lunar Gateway space station (NASA led)
	<i>CSP</i>	Communication services project for commercial space relay communication services (NASA led)
	<i>HLS (Human Landing System)</i>	Part of Artemis program to take astronauts to the moon (NASA led)
Commercial	<i>CLPS-1 (Commercial Lunar Payload Services, Task 1)</i>	Mission to deliver science and technology to the lunar surface (NASA led)
	<i>CLPS-2 (Commercial Lunar Payload Services, Task 2)</i>	Mission to deliver science and technology to the lunar surface (NASA led)

(continued on next page)

Table 3 (continued)

Space PPP category	Case	Brief description
	<i>Cleerspace-1</i>	Mission for space debris removal demonstration (ESA led)
	<i>Next Gen Spacesuits</i>	New spacesuit for NASA, for the ISS and lunar spacewalks (NASA led)

Table 4

Affiliations and profiles of the informants included in the sample.

Informant ID	Organisation	Informant Profile	Experience	Duration
Inf. #1	Space agency (Western)	Associate Administrator	21 years	58 min
Inf. #2	Space agency (Western)	Research Analyst	17 years	57 min
Inf. #3	Space agency (Western)	Senior System Engineer	22 years	64 min
Inf. #4	Space agency (Western)	Program Analyst	12 years	63 min
Inf. #5	Space agency (non-Western)	Senior Researcher	9 years	43 min
Inf. #6	Start up (Western)	Head of Communication	5 years	31 min
Inf. #7	Large enterprise (Western)	Chief Engineer	16 years	53 min
Inf. #8	Space research centre (Western)	System Engineer	22 years	43 min
Inf. #9	Space agency (non-Western)	System Engineer	12 years	34 min

the validity of findings, we triangulated data with publicly available documents, material provided by informants [44], follow-up questions with informants to address doubts [56], and member checking [54]. Table 5 exemplifies the multi-dimensional framework employed for the within-case and cross-case analyses. The framework enabled us to capture key points, including actors and their objectives/capabilities, project-specific variables, procurement design, contextual factors, challenges and risks. This framework was used to systematize case data, to support cross-case synthesis, and to identify recurring patterns and outliers.

Data from interviews and case coding were analysed using thematic analysis. Emerging themes were compared within and across PPP categories to surface key drivers and conditions for adoption. Particular attention was paid to divergent cases where procurement choices deviated from expected patterns.

4. Findings

4.1. Key drivers and conditions guiding the procurement practice selection

Below, we present the key drivers and conditions supporting the selection of the procurement practice for the project of interest. These key drivers and conditions discriminate whether a project should fall under the left side of the space PPP spectrum (i.e. conventional practices), the right side (i.e. commercial practices), or in between the two

Table 5

Framework for within-case and cross-case analyses.

Category	Characteristics
Space Agency Perspective	Agency name, Strategic goals, Capabilities, Market steering, Strategic autonomy, Risk/cost/time reduction
Procurement Design	Scope of work, Type of contract, Customer, #of bidders/awardees, Competitive phases, Space PPP category
Challenges and constraints	Budget, Time, Termination, Other (e.g., technical failures)

(following the space PPP typologies proposed by Kim (2023) [2]). The drivers and conditions were identified through an abductive process, combining a deductive review of PPP literature with inductive insights emerging from case study evidence. We first extracted recurring variables emphasized in prior PPP studies (e.g., technological maturity, demand, supply structure). Through case study analysis, we further refined and expanded the set of relevant variables taking advantage of themes emerging from divergent or hybrid cases that revealed additional drivers and conditions not fully captured by prior literature. Informants' perspectives provided an additional layer of validation, confirming the relevance of some drivers and conditions while introducing new ones (e.g., contractor experience).

From this process, we identified 10 key drivers and conditions guiding the selection of the most appropriate space PPP typology. We organized the final set of drivers and conditions into four categories.

1) Technological readiness and project maturity. This group comprises the Technology Readiness Level (TRL), the mission maturity, and the predicted length of the projects.

1.1) Technology Readiness Level (TRL): the TRL scale provides a measure of technological maturity spanning from TRL1, "basic principle" technology, to TRL9, "flight proven" technology [57]. Across cases and interviews, TRL was found to be a primary determinant in guiding procurement models. Lower TRLs are typically associated with conventional procurement approaches (e.g., cost-plus, agency-led development), while higher TRLs facilitate the adoption of commercially oriented partnerships. For example, the first Galileo satellites (Giove-A and Giove-B) were procured through an agency-driven approach at a time when key navigation payload technologies were at low TRL. On the contrary, high-TRL systems support performance-based and service-oriented contracts. NASA's Gateway Logistics Services (GLS) relies on the Dragon XL spacecraft, itself an upgrade of the flight-proven Dragon vehicle, enabling a fixed-price, commercially structured agreement. Informants #7 and #9 reinforced this logic "If there are technology development projects, so that they have a high technological innovation part, the participation is then close control, but also really the active participation of the agencies." (Inf. #7) "We give to companies freedom in the details management when the core technology is a well-developed long-term concept." (Inf. #9).

1.2) Mission Maturity: as with the TRL, the lower the maturity and the experience of contractors and space agencies on similar efforts (i.e., in case of brand new projects), the greater the responsibility and participation of the space agency (i.e. the more conventional the practice). Conversely, when the project involves the redesign, upgrade, or industrialisation of an already-tested technology, the private intervention and responsibility are higher, given the lower risk related to the technology maturity. Next-Gen OPIR introduced novel technologies and capabilities in a defence context and was managed through a conventional approach. Contrarily, Next Gen Spacesuits, at least for the ISS suit, were based on decades of EVA suit experience, with contractors building on established designs and operational lessons through service-driven contracts. As Informant #7 stated, "Then once it is a matter of getting a concept that has been tested, validated, to the next stage, meaning the industrialization stage, the responsibility curve starts to steepen, where the responsibility becomes more and more on the private side.".

1.3) Predicted length of the project: the predicted length of the contract also affects the choice of procurement practice. Longer projects and missions entail longer R&D cycles, demanding therefore conventional practices (due to lower TRL). Conversely, shorter projects (e.g. demonstrations and high TRL projects) translate into earlier potential returns and go-to-market, raising interest in private partners. Davinci mission, managed through a conventional approach, was a complex planetary science mission with extensive development cycles for instruments and spacecraft. On the other side of the spectrum, CSP uses existing servicing heritage, requiring shorter development for operational deployment and faster go-to-market for the companies involved. "If you look at the projects that fall on the left side, they are the longer-term

projects because they require a lot more R&D, and they require new things that are getting developed. So they will take seven, 8-9-10 years. The private sector doesn't have the investing appetite in them. A lot of these startups, a lot of these venture capitals, they are looking to generate returns within the next 5-6 years. The government is going to do the things that require more basic research, whereas the private sector would do the things that the government has already perfected and pushed out. The TRL level is very high, so consequently the ones on the right side would take shorter amount of time." (Inf. 5).

2) Market context and private sector involvement. This group comprises the demand maturity, supplier availability, type of partner, and contractor experience.

2.1) Demand maturity: it refers to the presence of a commercial market, where clients and users differ from the sole space agency guiding the program or project. A key element is whether private companies (other than space agencies) are interested in the product/service/technology, and potentially willing to buy them in the future. Thus, demand is the main avenue for private partners to achieve an ROI in the future, as well as to assume the risks, responsibilities, and costs associated with the PPP. The lower the demand, the lower the willingness of private players to engage in a PPP, contributing resources. The higher the demand, or the willingness of the space agency to create and signal it, the more commercial the procurement practices can be. ICESat-2 was a scientific mission with no commercial market, targeting climate research and Earth science. Conversely, COTS was based on NASA as anchor customer with recurrent resupply contracts, and CLPS had signalled potential future science and commercial customers. As two informants highlighted, "Telecom side is very commercial so the presence of the agencies is almost negligible, they collaborate, they give support, but overall they are not in the front line. It's different when we talk about programmes like Copernicus, which are commissioned by the agency and where the ultimate customer and the ultimate operator is the agency, the European Community." (Inf. 7) "Where the government owns and repairs the ownership, perhaps that's less important. Perhaps because those programs don't look for commercial markets. They are meant for government to own and operate. So while they might cost more, they have a different goal. The goal wasn't to reduce cost on those types of partnerships [...] who is willing to invest billions of dollars in that? Because that has a very high risk of not being successful or not being able to be commercialized. It's not just whether the government wants to push the technology, it's whether the private sector is willing to take on those risks as well, and this depends of whether they see a commercial market." (Inf. 5).

2.2) Available Suppliers: it refers to the availability of possible candidates and vendors that can concretely compete and deliver the product/service to the space agency. In cases where there is a limited number of available suppliers, the space agency exerts greater control over the project phases, thereby favouring conventional practices. However, this situation can pose a challenge for the space agency, as the sole available supplier possesses significant bargaining power, potentially leading to a lock-in scenario for the agency. Conversely, if the supply pool is robust, the procurement process may exhibit greater competitiveness and lean towards commercial practices. Next-Gen OPIR could initially rely on a limited sample US aerospace primes, while programs like COTS and CLPS still relied on a limited number of capable companies, but with NASA running open competition to stimulate entrants along multiple program phases. "To have a very successful partnership, the demand has to be there, but also it has to have robust supply side. If we're creating a monopoly, they are going to act monopolistic and we will take the hit on the monopoly side. [...] The demand pool and the supply pool have to be there." (Inf. 5).

2.3) Contractor experience: contractors may have experience in past similar projects or missions, or lacking background experience. This influences the choice of the procurement practice, as with experience contractors may be in the position of bearing more responsibilities. Without previous experience in similar efforts, contractors need the support, both technical and financial, of the space agency. At program

inception in the late '70s, industrial partners involved in SPOT 1-2-3 had no EO commercial satellite experience, and CNES led the architecture and payload definition. On the contrary, in MRV (RSGS) Northrop Grumman could leverage MEV servicing heritage for the MRV bus. As informants pointed out; "Space agencies give contracts to companies that have superior skills than those of the agency." (Inf. 3) "It is an open competition, so you have to show your competences against the target. Showing competences is based on previous programmes, having done things like that before. It's based on your skills, in the sense of saying, actually I have internal skills to do the development, I don't know, hardware, software, or manpower, business skills" (Inf. 7).

2.4) Type of private partner: in the case of startups and small companies, the role of the space agency is still essential, as the private partner lacks some knowledge, capabilities, but also resources (e.g. facilities, and personnel). On the contrary, large enterprises (e.g. incumbents) might possess the needed software, capabilities, and human resources to internally manage projects. In the BEAM project, Bigelow Aerospace was a small private company with no prior human-rated spaceflight experience, and NASA handled safety, integration, and ISS interface management. On the other side, the HLS involves major primes like SpaceX and Blue Origin, both with extensive development resources. Some interviewees reinforced this point; "An agency usually avoids giving too big contracts to a small company that has not already demonstrated the competence to complete the contract." (Inf. 3) "Depending on the size of the company, the role and importance of collaboration with the space agency is completely different." (Inf. 7)

3) Strategic importance. This group comprises project objective and project importance.

3.1) Objective of the project: across cases and informants, the objective of the project is distinguished between "living and operating in space" and "doing science in space". Missions in the former category are repetitive; in these missions, private sector participation is encouraged, as costs and risks tend to diminish over time due to the iterative nature of the processes, which become refined and efficient. Additionally, the repetition of missions generates future cash flows, thereby supporting ROI and establishing a commercial demand base. Examples include the GLS, the HLS program, and CSP, all of which leverage private-sector capabilities for recurring operational activities. Conversely, the second category entails one-off missions, aimed at the achievement of scientific objectives. A commercial market for these missions is hardly present, demanding therefore the space agency to bear the highest share of risks and costs. These projects, such as Davinci, ICESat-2, and Maven, rarely have a viable commercial market beyond space agencies and governmental entities, making service-driven procurement challenging. As one informant explained; "Living and operating in space will be the commercial LEO destinations, space stations, lunar landing launch vehicles and space communications. And they are moving towards that service driven contracts where NASA is just buying services from them, from companies, for example. And these are usually done as what's called firm fixed price contracts. [...] So science missions are generally just one-off missions. We don't repeat things in science. And it wouldn't make sense for these to be a service driven contracts or procurements, because if space agencies and a few government entities are the only clients for these, what is the point? There's no commercial market there." (Inf. 5).

3.2) Size and importance of the project: another determinant in procurement practice selection is the strategic weight of the project. Flagship programs tend to feature stronger agency participation, tighter oversight, and reduced reliance on commercial partners, given the consequences of potential failure. Examples include Next-Gen OPIR, and Galileo (1st generation), where agency-led or closely supervised procurement mitigated the risks of project disruption. In contrast, secondary projects, while still technically sophisticated, allow greater private-sector participation, cost-sharing, and risk assumption. Cases such as BEAM, Clearspace-1, and certain CLPS payload deliveries illustrate how agencies adopt more commercially oriented practices when mission-criticality is lower. "The commercial company is not legally or

contractually bound to deliver. Of course, if they don't deliver, they don't get the money. It's that simple. But then what happens? The entire NASA Artemis strategy comes to an absolute standstill. It is a single point of complete mission failure that we are contracting out fully commercially. The same goes for commercial low Earth orbit destinations. [...] We have to weigh the inherent commercial risks of doing things a certain way with how important that project is to achieving key mission elements" (Inf. 4).

4) Requirements and solution definition.

When the solution and technical requirements are clearly defined upfront, before starting the procurement process, space agencies tend to follow a "hands-on" approach to the project, favouring conventional practices and frequent reviews to check compliance with specifications. For example, in missions like Next-Gen OPIR or the first Galileo satellites, the agency specified detailed subsystem characteristics, leaving limited design freedom to contractors. On the contrary, commercial procurement models often rely on a hands-off approach, in which the agency specifies *what* performance or capability is required, while leaving the *how* of the specific solution developed to the private partner. This approach is common in service-based contracts, where outcomes matter more than the asset configuration. Certain CLPS missions illustrate this approach, alongside the HLS, granting contractors autonomy in integration decisions to meet high-level performance objectives. As one informant summarised; *"The logic behind this thing is that selling a service, or rather, making a contract to acquire a service requires fewer constraints in the contract itself. If I want to buy a specific satellite I say exactly what I want. A satellite is a particularly complex object, saying I want a satellite means nothing, you really have to go into detail. If we want to buy a service, it is simpler to specify a service and let the start-up decide how to implement its hardware to meet a need."* (Inf. 3).

Table 6 summarizes the key drivers and conditions.

4.2. Recurrent patterns and outliers in the selection logic of procurement practices

Some projects and programs are managed in line with the practices typical of their PPP category, with procurement choices adapting as shifts in key drivers and conditions influence the selection of practices, as illustrated in Fig. 2. Yet, not all of their underlying drivers and conditions are fully aligned with those that the category would theoretically be expected to display. Certain drivers and conditions can outweigh

others; for example, strategic autonomy or policy mandates may override high TRL or supplier availability, leading to a conventional approach even when commercial indicators are present. Conversely, some projects are deliberately steered toward a commercial orientation despite many drivers and conditions are aligned with conditions for adopting conventional practices.

Examples from the cases illustrate this dynamic. In the Conventional category, Next Gen OPIR combines "live and operate in space" capabilities with high contractor experience, both typical of commercial-oriented projects; yet, defence imperatives oriented the agency toward a fully conventional model. On the other hand, certain CLPS projects adopted a commercial framework despite low TRL, inexperienced contractors, new asset concepts, and only "foreseeable" future demand, reflecting NASA's strategic choice to stimulate a nascent lunar delivery market. Empirical evidence and the discrepancies between the theoretical space PPP guidelines and the implemented procurement practices indicate the necessity for more thorough analysis and discussion.

5. Discussion

5.1. Convergencies between space and non-space PPPs

Several drivers and conditions shaping procurement practices in the space sector are consistent with those identified in non-space PPP literature.

Demand maturity is a central determinant of PPP feasibility, as projects without clear revenue stream struggle to attract private participation [11–13]. Space PPPs align closely; missions with limited commercial market (e.g., ICESat-2) remains agency-led, while those with recurrent demand (e.g., COTS, CLPS) allow greater risk transfer through the adoption of commercial-oriented forms of PPPs.

Strategic importance, in terms of economic weight and size, alongside political visibility of projects usually feature stronger institutional oversight due to high risks and political sensitivity (e.g., high-speed rail, nuclear plants) [14,19,20]. Procurement practices in the space sector mirror this dynamic; Galileo satellites and Next-Gen OPIR stayed agency-driven due to strategic weight, while lower-stakes projects (e.g., BEAM) saw more commercial involvement.

In non-space PPP literature, a competitive supply base is essential to ensure Value for Money (VfM); lack of competition risks indeed

Table 6
Space PPPs drivers and conditions.

THEMATIC BLOCK	DRIVER	DEFINITION	CONVENTIONAL	COMMERCIAL	EXAMPLE – CONVENTIONAL	EXAMPLE - COMMERCIAL
Technological readiness and project maturity	TRL	Maturity of technology from TRL1 to TRL9	Low TRL -> agency-led	High TRL -> service-based	Galileo Giove-A and Giove-B satellites	Gateway Logistics Services
	Mission maturity	Extent of heritage/prior similar missions	Brand new-> high agency involvement	Upgrade/industrialisation -> high private role	Next-Gen OPIR	Next-Gen Spacesuits
	Predicted length	Expected R&D/project duration	Long -> conventional (due to low TRL)	Short -> private interest, faster ROI	Davinci	CSP
Market context and private sector involvement	Demand maturity	Presence of commercial market	No demand -> conventional	High/signalled demand -> commercial	ICESat-2	CLPS
	Available suppliers	Number of qualified suppliers	Limited suppliers -> agency control	Medium/high suppliers -> competition	Next-Gen OPIR	COTS
	Contractor experience	Past similar project experience	Low -> agency support	High -> more private responsibility	SPOT 1-2-3	MRV(RSGS)
Strategic importance	Type of private partner	Size and capabilities of partner	Startups/SMEs -> more agency support	Large primes -> more autonomy	BEAM	HLS
	Project objective	"Science" vs "Live and operate in space"	Science -> one-off, agency-led	Operate/living in space -> recurring revenues, service-based	Maven	GLS
	Project importance	Flagship vs secondary	Flagship -> hands-on	Secondary -> hands-off	Galileo 1st gen. satellites	Some CLPS payload deliveries
Requirements and solution definition	Requirements and solution definition	Degree of predefined technical aspects	Fully defined -> hands-on	Performance-defined -> hands-off	Next-Gen OPIR	HLS

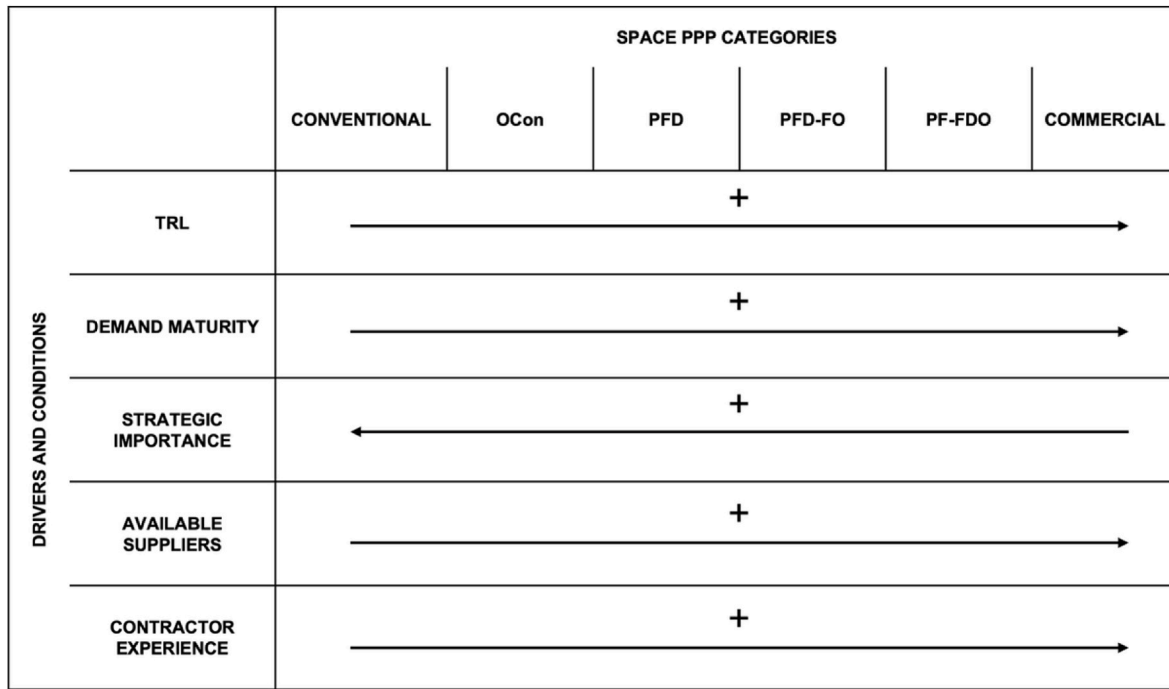


Fig. 2. Drivers and conditions influencing procurement choices by space agencies.

monopolies and lock-ins [13,15,58]. In the space sector, a similar logic holds true, but with sharper effects due to sometimes limited pools of space primes; NASA often runs open competitions (e.g., COTS, CLPS) to expand suppliers and avoid dependency [59–62].

Last, contractor experience shapes risk allocation and procurement choices in non-space PPPs; inexperienced contractors usually require more public oversight [12,16]. The space sector is strongly consistent with this logic; startups like Bigelow (in BEAM project) required NASA's deep involvement, while SpaceX and Blue Origin, once involved in the HLS, could assume greater responsibility and degree of autonomy in solution design decisions.

5.2. Space-specific drivers and conditions of PPPs

At the same time, our findings highlight drivers and conditions that are peculiar to the space domain and not specifically addressed by non-space PPP literature.

A central nuance lies in the TRL scale. While technological maturity is acknowledged conceptually in non-space literature (e.g., risks in innovative infrastructures), there is no standardised and operational measure in general PPP research. In the space domain, the TRL provides a precise and institutionalized scale (TRL 1–9), directly influencing procurement choices. Projects at low TRLs tend to require conventional models, as exemplified by Giove-A and Giove-B satellites within Galileo Program. Conversely, higher TRLs reduce technological risk and open the way for commercially oriented approaches. This leads to the formulation of:

Proposition 1. *Higher TRLs increase the likelihood that procurement practices shift toward commercially oriented PPP typologies.*

Additionally, in non-space literature, “strategic importance” typically points to the economic weight, size, and political visibility of projects, and is linked to national prestige and risk of failure, which drive governments to retain stronger control [14,19]. In the space sector, however, strategic importance does highlight the political visibility of projects (e.g., the Apollo program for U.S. leadership in space exploration) but also encompasses a relevant Security&Defence dimension, largely absent or peripheral in other types of infrastructure

systems with limited influence on the PPP arrangements. Programs like Next-Gen OPIR carry direct military and security functions. Even civil missions (e.g., Galileo program) often have dual-use applications blending civilian, security and defence objectives. This means that strategic importance in space is not only about flagship size and visibility, but also geopolitical and defence-related stakes, which strongly influence PPP choices and make governments less willing to transfer responsibility or risk to private partners. This leads to the formulation of:

Proposition 2. *The higher the (strategic) Security&Defence importance of an initiative, the more procurement practices remain anchored in conventional models.*

Last, in non-space PPPs, demand maturity is usually observable, measurable, and market-driven (e.g., electricity consumption). PPP is judged by whether there is sufficient existing or forecastable private demand to ensure ROI [11–13]. In the space sector, commercial demand is often absent or emergent. Space agencies therefore actively create, signal, or anchor demand to de-risk private investment. This takes form such as anchor tenancy (e.g., in NASA's COTS) and long-term commitments (e.g., in CLPS where NASA announced recurring lunar delivery opportunities), a practice rarely seen in other domains. This makes “demand maturity” in space less about an existing market and more about agency-enabled or agency-signalled demand. This dynamic directly influences procurement choices; the stronger and more credible the demand, whether market-driven or agency-signalled, the more feasible and attractive commercially structured practices become. This leads to the formulation of:

Proposition 3. *The Space Agency (the buyer) plays a key role in fostering higher demand maturity, actively favouring the adoption of commercially-structured procurement practices.*

5.3. Joint effects of drivers and conditions in space PPPs

The findings also indicate that drivers and conditions in the space sector rarely operate in isolation; rather, their effects often intersect, producing outcomes that deviate from what single variables would predict. This recognition is well-established in the non-space PPP

literature [63–68], though it remains limitedly theorised in the space PPP context.

As an example, programs/projects may be managed through conventional approaches when strategic and defence relevance prevail, even in cases characterised by high TRL, experienced contractors, and recurrent demand (conditions that would typically support commercially-oriented procurement). Next-Gen OPIR and SPOT 1-2-3 satellites illustrate this logic; despite favourable technological (e.g., SPOT), market conditions (e.g., EO demand for SPOT, “live and operate in space” for Next-Gen OPIR) and experienced contractors (e.g., Next-Gen OPIR), procurement was kept under close agency control to ensure full oversight of system design, development, and integration, thereby safeguarding security and geopolitical leadership. This leads to the formulation of:

Proposition 4. *Even when high TRL, contractor experience, and recurrent demand are present, initiatives of high strategic or defence relevance tend to be developed under conventional procurement.*

Conversely, the opposite scenario can also occur. Projects with low TRL and a limited supplier base would normally suggest adopting conventional procurement, yet when trusted contractors with prior contracting experience with the space agency are involved, more risk sharing and responsibility can be transferred to them. The case of SLIM with Mitsubishi exemplifies this logic, where JAXA leveraged its long-standing relationship and confidence in the contractor’s capabilities to relax control. This leads to the formulation of:

Proposition 5. *In cases of low TRL and limited supplier base, the space agency can still enable risk-sharing and commercially oriented practices when contractors with established trust and extensive past experience are involved.*

6. Conclusions

This study demonstrates that procurement practices in the space sector cannot be understood through a simple transfer of general PPP adoption guidelines. While several drivers and conditions, such as technology maturity, demand maturity, supplier diversity, and contractor experience, are consistent with and confirm indications from non-space PPP literature, our study highlights how their interaction, and the distinctive features of the space domain, make procurement choices relatively more challenging. Space-specific nuances include the industry-wide use of a standard TRL scale, the role of space agencies in actively signalling or anchoring demand, and the Security&Defence and geopolitical stakes that often outweigh favourable technological, market, or supply base conditions. These dynamics show that drivers and conditions rarely operate in isolation but interact in ways that may either reinforce commercial procurement logics or maintain conventional models despite favourable conditions for greater risk-transfer.

From our study, three main insights emerge. First, higher TRL and contractor experience generally increase the likelihood of shifting toward commercially oriented procurement. Second, greater demand maturity, whether market-driven or agency-signalled, combined with supplier availability supports the adoption of commercial practices. Third, when projects are of high strategic or Security&Defence relevance, conventional procurement prevails regardless of technological or market readiness. At the same time, trusted relationships between the space agency and experienced contractors can enable risk-sharing even when TRL is low.

Several recommendations for space agencies emerge. Space agencies should adopt dynamic procurement models. They should systematically assess TRL, demand maturity, supplier base, and contractor experience before awarding contracts, and adapt procurement practices as these conditions evolve rather than locking them for the entire duration of the programs. Space agencies should balance commercialisation with strategic control. Commercial procurement should not be adopted for political visibility or innovation branding alone. When projects are

strongly linked to Security&Defence objectives or are strategically sensitive, conventional practices with stronger public oversight remain the best option. Last, they should structure PPPs in emerging sectors (e.g., in-orbit servicing, ADR) with staged de-risking. Where demand is weak or only foreseeable, agencies should pair strong public incentives with gradual transfer of responsibility to industry. This includes avoiding premature full commercialisation, and monitoring and continuously communicating market conditions.

The qualitative evidence and theory-informed case analysis suggested several testable propositions. Future research should quantitatively test these propositions across a broader sample of PPP programs, controlling for variables such as national context, agency maturity, and industrial base strength. Furthermore, future research should delve deeper into specific procurement aspects such as the tendering process, milestones, business models, and price and cost assumptions. We recommend examining projects with similar attributes to facilitate comparative studies. Finally, further work should examine procurement sub-tools and hybrid logics, such as parallel developments and ecosystem approaches, as well as the economics of fixed-price contracting for innovations that do not yet exist, including how prices are set when market comparables are absent.

This study focuses on large, established space agencies. We did not cover smaller or emerging space agencies (by design) to capture the transition dynamics of mature actors. Future work should investigate how younger agencies, such as ISRO or other newer entrants, approach commercial procurement from the outset, including how they build industrial capabilities that established agencies already possess.

Defence-related programs were addressed only marginally. While the analysis identified the role of geopolitical tensions and objectives and their influence on procurement models, more granular, defence-focused case studies are needed to understand how security imperatives shape PPP adoption, risk-sharing, and contractor engagement.

CRedit authorship contribution statement

Valentina Zancan: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Andrea Pozzi:** Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alessandro Rota:** Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sascha Deutsch:** Writing – original draft, Visualization, Validation, Supervision, Project administration, Formal analysis, Conceptualization. **Paolo Trucco:** Writing – original draft, Visualization, Validation, Project administration, Methodology, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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